Experiment 15: EasyToll – Online Toll Payment System

1. Aim

To design and develop an interactive online toll payment system named EasyToll using HTML, CSS, and JavaScript that allows users to manage toll payments, add money to an eWallet, purchase advance toll eTickets, apply for toll passes, and generate QR codes for toll gate verification.

2. Procedure

1. Create the website structure using HTML.
2. Design a responsive user interface using CSS.
3. Implement user registration and login functionality using JavaScript.
4. Develop an eWallet module to add money and display the available balance.
5. Implement vehicle registration and vehicle detail management.
6. Create a toll eTicket purchasing system that allows users to select a toll gate, vehicle, travel date, and time.
7. Develop toll pass application and renewal features.
8. Implement administrator functions to manage users, approve or reject toll pass applications, and create, edit, and delete toll gate details.
9. Generate demo QR codes for toll tickets and approved toll passes.
10. Develop a toll gate staff portal to verify ticket and pass IDs and update toll gate locations.
11. Store user information, tickets, passes, and transaction records using browser localStorage.
12. Test all modules using a web browser.

3. Code Snippet

HTML – Buy Toll eTicket Button

```
<button onclick="buyTicket()">
   Buy Toll eTicket
</button>
```

JavaScript – Add Money to eWallet

```
function addMoney(event) {
   event.preventDefault();


   let amount = Number(
```

```
    document.getElementById("addAmount").value
  );

  if (amount < 100 || amount > 50000) {
    toast("Enter a valid amount");
    return;
  }

  user.balance = Number(user.balance || 0) + amount;

  db.transactions.unshift({
    id: "TX" + Date.now(),
    owner: user.email,
    type: "Wallet top-up",
    amount: amount,
    status: "Success",
    date: new Date().toISOString()
  });

  save();
  toast("Wallet balance updated");
}
```

JavaScript – Verify Toll Pass or eTicket

```
function verifyCode(event) {
  event.preventDefault();

  let id = document.getElementById("verifyId")
    .value.trim().toUpperCase();

  let pass = db.passes.find(
```

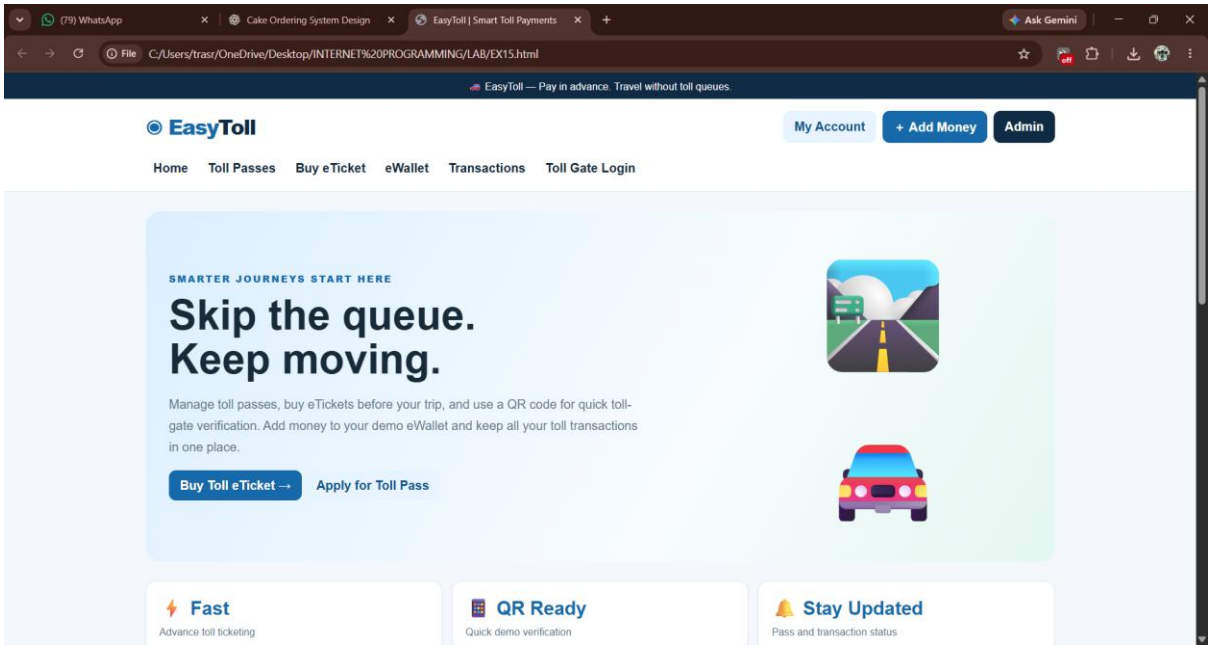
```
        p => p.id.toUpperCase() === id
    );

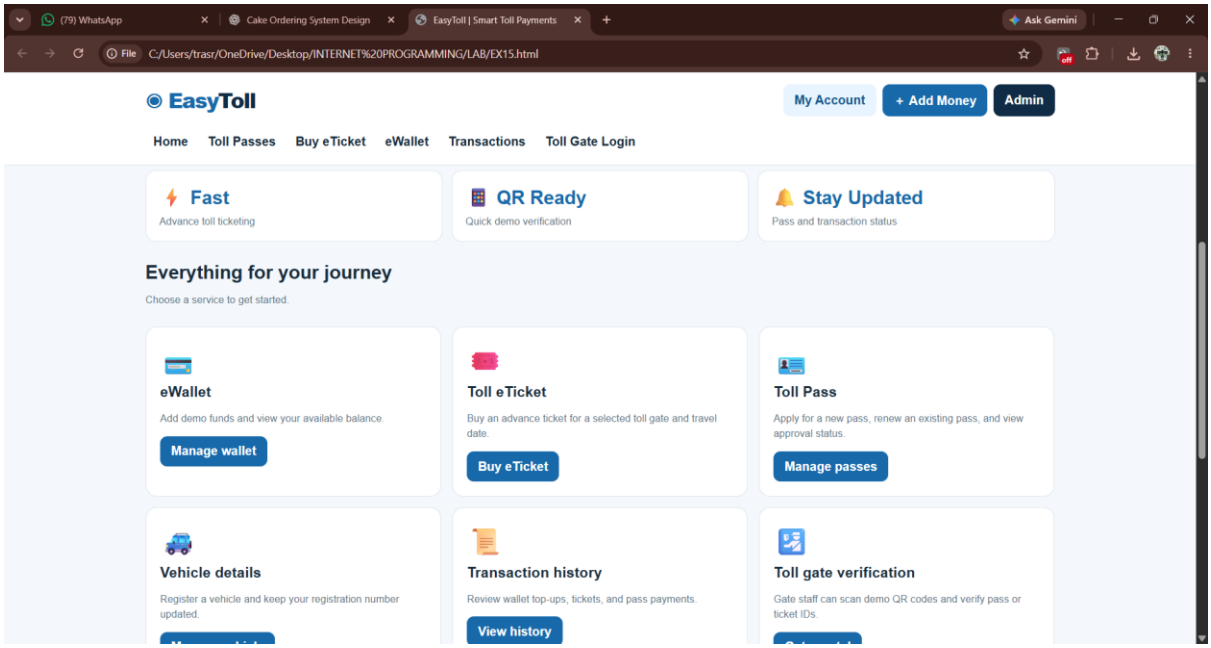
    let ticket = db.tickets.find(
        t => t.id.toUpperCase() === id
    );

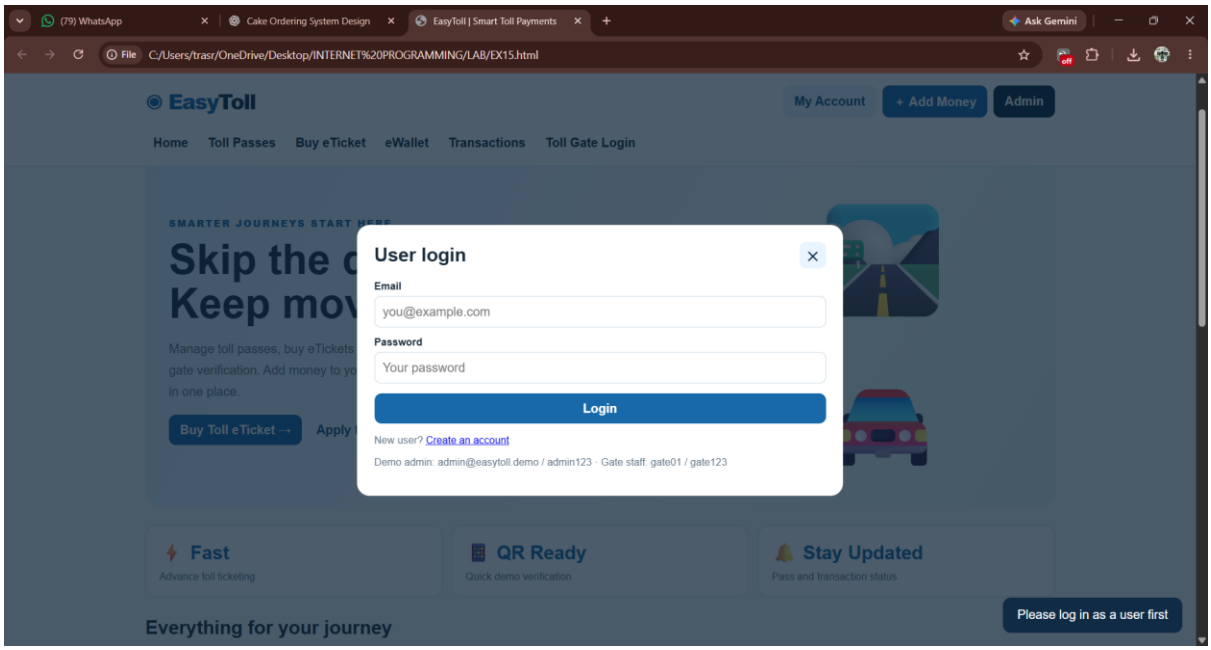
    let result = "Invalid";

    if (pass && (
        pass.status === "Active" ||
        pass.status === "Approved"
    )) {
        result = "Valid";
    } else if (ticket && ticket.status === "Valid") {
        result = "Valid";
    }
    toast("Verification complete: " + result);
}
```

4. Output







5. Result

Thus, the EasyToll online toll payment system was successfully designed and developed using HTML, CSS, and JavaScript. The system demonstrates advance toll ticket booking, eWallet management, toll pass approval, QR-based verification, and administrative management through a user-friendly interface.